Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The solubility product $(\mathrm{K_{sp}})$ of $\mathrm{Fe(OH)_3}$ aqueous solution is 3.8×10^{-38} at $298\ \mathrm{K}$. The solubility of $\mathrm{Fe^{3+}}$ ions will increase when

(A) Saturated solution is exposed to atmosphere

(B) pH is 7

(C) pH is decreased

(D) pH is increased

Q.2

$\mathrm{Ag_2CrO_4}$ and $\mathrm{Ag_2CO_3}$ are simultaneously dissolved in water. The solubility of $\mathrm{Ag_2CrO_4}\&$ $\mathrm{Ag_2CO_3}$ are x' and $'y'$ mo M respectively. If the $\mathrm{K_{ep}}$ of $\mathrm{Ag_2CrO_4}$ is 'a' and that of $\mathrm{Ag_2CO_3}$ is 'b', then which of the following is correct?

(A) $(x+y)^2x=n$

(B) $(2x+2y)y=b$

(C) $(2x+2y)^2x=b$

(D) $(2y+2x)^2y=b$

Q.3

The $\mathrm{K_{sp}}$ of silver carbonate is $5.0 \times 10^{-13}\mathrm{M^3}$. Hence solubility of silver carbonate $(\mathrm{Mw}=276)$ in pure water is

(A) $13.80\mathrm{ppm}$

(B) $21.9\mathrm{ppm}$

(C) $6.40\mathrm{ppm}$

(D) $10.95\mathrm{ppm}$

Q.4

Which is the most soluble from the following

(A) $\mathrm{Bi_2\ S_3}\left[\mathrm{K_{ep}}=1\times10^{-70}\right]$

(B) $\mathrm{MnS}\left[\mathrm{K_{sp}}=7\times10^{-16}\right.$

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(C) $\mathrm{CuS}\left[\mathrm{K_{ap}} = 8 \times 10^{-37}\right]$

(D) $\mathrm{Ag_2\ S}\left[\ \mathrm{K_{sp}} = 6 \times 10^{-51}\right]$

Q.5

If K_{sp} of $\mathrm{CaF_2}$ at $25^{\circ}\mathrm{C}$ is 1.7×10^{-10}, the combination amongst the following which gives precipitate of $\mathrm{CaF_2}$ is

(A) $1 \times 10^{-2}\mathrm{MCa^{2+}}$ and $1 \times 10^{-3}\mathrm{MF^{-}}$

(B) $1 \times 10^{-4}\mathrm{MCa^{2+}}$ and $1 \times 10^{-4}\mathrm{MF^{-}}$

(C) $1 \times 10^{-2}\mathrm{MCa^{2+}}$ and $1 \times 10^{-5}\mathrm{MF^{-}}$

(D) $1 \times 10^{-3}\mathrm{MCa^{2+}}$ and $1 \times 10^{-5}\mathrm{MF^{-}}$

Q.6

The solubility of $\mathrm{PbI_2}$ at $25^{\circ}\mathrm{C}$ is $0.7\ \mathrm{g\ L^{-1}}$. The solubility product of $\mathrm{PbI_2}$ at this temperature is (molar mass of $\mathrm{PbI_2} = 461.2\ \mathrm{g\ mol^{-1}}$)

(A) 1.40×10^{-9}

(B) 0.14×10^{-9}

(C) 140×10^{-9}

(D) 14.0×10^{-9}

Q.7

The solubility product of $\mathrm{Ag_2CrO_4}$ is 32×10^{-12}.
What is the concentration of $\mathrm{CrO_4^{2-}}$ ions in that solution (in $\mathrm{gL^{-1}}$)

(A) 2×10^{-4}

(B) 8×10^{-4}

(C) 8×10^{-8}

(D) 16×10^{-4}

Q.8

What is the minimum concentration of $\mathrm{SO_4^2}$ required to precipitate $\mathrm{BaSO_4}$ in a solution containing 1.0×10^{-4} mole of $\mathrm{Ba^{2+}}$? $\mathrm{K_{sp}}$ for $\mathrm{BaSO_4} = 4 \times 10^{-10}$

(A) $4 \times 10^{-10}\mathrm{M}$

(B) $2 \times 10^{-7}\mathrm{M}$

(C) $4 \times 10^{-6}\mathrm{M}$

(D) $2 \times 10^{-3}\mathrm{M}$

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Q.9

Consider the following equilibrium $\mathrm{AgCl} \downarrow +2\mathrm{NH_3} \rightleftharpoons \left[\mathrm{Ag}\left(\mathrm{NH_3}\right)_2\right]^+ + \mathrm{Cl}^-$
White precipitate of AgCl appears on adding which of the following?

(A) $\mathrm{NH_3}$

(B) aqueous NaCl

(C) aqueous $\mathrm{HNO_3}$

(D) aqueous $\mathrm{NH_4Cl}$

Q.10

A litre of solution is saturated with AgCl. To this solution if 1.0×10^{-4} mole of solid NaCl is added, what will be the $\left[\mathrm{Ag}^+\right]$, assuming no volume change?

(A) More

(B) Less

(C) Equal

(D) Zero

Q.11

How many gms of $\mathrm{CaC_2O_4}$ will dissolve in one litre of saturated solution. $\mathrm{K_{sp}}$ of $\mathrm{CaC_2O_4}$ is $2.5 \times 10^{-9}\ \mathrm{mol^2 lit^{-2}}$

(A) 0.0064 g

(B) 0.0128 g

(C) 0.0032 g

(D) None of these

Q.12

Zirconium phosphate $\left[\mathrm{Zr_3}\left(\mathrm{PO_4}\right)_4\right]$ dissociates into three zirconium cations of charge $+4$ and four phosphate anions of charge -3. If molar solubility of zirconium phosphate is denoted by S and its solubility product by $\mathrm{K_{sp}}$ then which of the following relationship between S and $\mathrm{K_{sp}}$ is correct?

(A) $\mathrm{S} = \left\{\mathrm{K_{sp}}/(6912)^{1/7}\right\}$

(B) $\mathrm{S} = \left\{\mathrm{K_{sp}}/144\right\}^{1/7}$

(C) $\mathrm{S} = \left\{\mathrm{K_{sp}}/6912\right\}^{1/7}$

(D) $\mathrm{S} = \left\{\mathrm{K_{sp}}/6912\right\}^{7}$

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Q.13

Solid $\mathrm{Ba}\left(\mathrm{NO}_3\right)_2$ is gradually dissolved in a $1.0 \times 10^{-4} \mathrm{MNa}_2 \mathrm{CO}_3$ solution. At which concentration of Ba^{2+}, precipitate of BaCO_3 begins to form ? $\left(K_{sp}\right.$ for $\left.\mathrm{BaCO}_3=5.1 \times 10^{-9}\right)$

(A) $5.1 \times 10^{-5} \mathrm{M}$

(B) $7.1 \times 10^{-8} \mathrm{M}$

(C) $4.1 \times 10^{-5} \mathrm{M}$

(D) $8.1 \times 10^{-7} \mathrm{M}$

Q.14

On addition of increasing amount of AgNO_3 to $0.1 \mathrm{M}$ each of NaCl and NaBr in a solution, what $\%$ of Br^{-} ion get precipitated when Cl^{-} ion starts precipitating. $\mathrm{K}_{12 \mathrm{sp}}(\mathrm{AgCl})=1.0 \times 10^{-10}$, $\mathrm{K}_{\mathrm{sp}}(\mathrm{AgBr})=1 \times 10^{-13} \mathrm{Pp}$

(A) 0.11

(B) 99.9

(C) 0.01

(D) 9.99

Q.15

Which one of the following arrangements represents the correct order of solubilities of sparingly soluble salts $\mathrm{Hg}_2 \mathrm{Cl}_2, \mathrm{Cr}_2\left(\mathrm{SO}_4\right)_3, \mathrm{BaSO}_4$ and CrCl_3 respectively?

(A) $\mathrm{BaSO}_4>\mathrm{Hg}_2 \mathrm{Cl}_2>\mathrm{Cr}_2\left(\mathrm{SO}_4\right)_3>\mathrm{CrCl}_3$

(B) $\mathrm{BaSO}_4>\mathrm{Hg}_2 \mathrm{Cl}_2>\mathrm{CrCl}_3>\mathrm{Cr}_2\left(\mathrm{SO}_4\right)_3$

(C) $\mathrm{BaSO}_4>\mathrm{CrCl}_3>\mathrm{Hg}_2 \mathrm{Cl}_2>\mathrm{Cr}_2\left(\mathrm{SO}_4\right)_3$

(D) $\mathrm{Hg}_2 \mathrm{Cl}_2>\mathrm{BaSO}_4>\mathrm{CrCl}_3>\mathrm{Cr}_2\left(\mathrm{SO}_4\right)_3$

Answers & Solutions

Q.1 Answer:

pH is decreased

Solution:

$\left[\mathrm{Fe}^{3+}\right]\left[\mathrm{OH}^{-1}\right]^3=\mathbf{K}_{\text{pp-}}$ **To increase** $\left[\mathrm{Fe}^{31}\right]$**, [OH] should decrease or [H'] will increase i.e.,** pH **should decrease.**

Q.2 Answer:

$(2y+2x)^2 y=b$

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Solution:

$\mathrm{Ag_2CrO_4(\ s) \rightleftharpoons 2Ag^+ + CrO_4^{2-}}$
$(2x + 2y) \quad x$
$\mathrm{Ag_2CO_3(\ s) \rightleftharpoons 2Ag^+ + CO_3^{2-}}$
$(2x + 2y)y$
$\mathrm{K_{ap}}$ **of** $\mathrm{Ag_2CrO_4 = a = (2x + 2y)^2 x}$
$\mathrm{K_{gp}}$ **of** $\mathrm{Ag_2CO_3 = b = (2x + 2y)^2 y}$

Q.3 Answer:

$13.80\mathrm{ppm}$

Solution:

$\mathrm{Ag_2CO_{3(tg} \rightleftharpoons 2Ag^+ + CO_3^{2-}}$
2s s
$\mathbf{K}_{-p} = 4\ \mathrm{s}^3$
$\mathrm{s} = \sqrt[3]{\frac{\mathrm{K_{gT}}}{4}} = \sqrt[3]{\frac{5 \times 10^{-13}}{4}} = 5 \times 10^{-5}\mathrm{M}$
$= 10^{-5} \times 276 = 1.38 \times 10^{-2}\ \mathrm{g/L} 10^3\ \mathrm{g}$ **water (or solution) contains** $1.38 \times 10^{-2}\ \mathrm{g Ag_2CO}$
$\therefore 10^6\ \mathrm{g}$ **water (or solution) contains** $13.8\ \mathrm{g Ag_2CO_3}$
Solubility $= 13.8$ **ppm**

Q.4 Answer:

$\mathrm{MnS}\left[\mathrm{K_{sp}} = 7 \times 10^{-16}\right.$

Solution:

Solubility is directly related with solubility product hence as K_m **increases solubility also increases.**
Most soluble salt is MnS.
$\mathrm{MnS} = \sqrt{K_{\mathrm{sp}}} = \sqrt{7 \times 10^{-16}} = 2.5 \times 10^{-3}$

Q.5 Answer:

$1 \times 10^{-2}\mathrm{M Ca^{2+}}$ and $1 \times 10^{-3}\mathrm{MF^-}$

Solution:

When ionic product i.e. the product of the concentration of ions in the solution exceeds the value of solubility product, formation of precitpiate occurs.
$\mathrm{CaF_2 \rightleftharpoons Ca^{2+} + 2\ F^-}$
Ionic product $= \left[\mathrm{Ca^{2+}}\right]\left[\mathrm{F^-}\right]^2$
when, $\left[\mathrm{Ca^{2+}}\right] = 1 \times 10^{-2}\mathrm{M}$
$\left[\mathrm{F^-}\right]^2 = \left(1 \times 10^{-3}\right)^2\ \mathrm{M}$
$= 1 \times 10^{-6}\mathrm{M}$
$\therefore \left[\mathrm{Ca^{2+}}\right]\left[\mathrm{F^-}\right]^2 = \left(1 \times 10^{-2}\right)\left(1 \times 10^{-6}\right) = 1 \times 10^{-8}$

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In this case, Ionic product $\left(1 \times 10^{-8}\right) >$ **solubility product** $\left(17 \times 10^{-10}\right)$

Q.6 Answer:

14.0×10^{-9}

Solution:

$\mathrm{PbI_2} \longrightarrow \mathrm{Pb_S^{++}} + 2\mathrm{I^-}$

$K_{sp} = s \times (2s)^2 = 4s^3$

$= 4 \times \left(\frac{0.7}{461.2}\right)^3 = 14.0 \times 10^{-9}$

Q.7 Answer:

2×10^{-4}

Solution:

For ternary electrolyte; solubility

$= \left(\frac{\mathrm{K_{sp}}}{4}\right)^{\frac{1}{3}} = \left(\frac{32 \times 10^{-12}}{4}\right)^{\frac{1}{3}} = 2 \times 10^{-4}$

Q.8 Answer:

$4 \times 10^{-6}\mathrm{M}$

Solution:

$\left[\mathrm{Ba^{2+}}\right]\left[\mathrm{SO_4^{2-}}\right]$

$\mathrm{K_{sp}} = \left(1.0 \times 10^{-4}\right)\left[\mathrm{SO_4^{2-}}\right] = 4 \times 10^{-10}$

$\therefore \left[\mathrm{SO_4^{2-}}\right] = 4 \times 10^{-6}\mathrm{M}$

Q.9 Answer:

aqueous $\mathrm{HNO_3}$

Solution:

$2\mathrm{HNO_3(aq)} + \left[\mathrm{Ag\,(NH_3)_2}\right]^+ + \mathrm{Cl^-}$

$\longrightarrow \mathrm{AgCl(s)} \downarrow + 2\mathrm{NH_4} + 2\mathrm{NO_3^-}$

When nitric acid is added to amine solution, solution is made acidic and the complex ion dissociates and liberate silver ion to recombine with chloride ion. This is the conformatory test for silver in group 1

Q.10 Answer:

Less

Solution:

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The dissociation of a weak electrolyte (AgCl) is suppressed on adding the strong electrolyte having a common ion $\left(Cl^{-}\right)$, hence concentration of Ag^{+} ion will be less.

Q.11 Answer:

0.0064 g

Solution:

$CaC_2O_4 \rightleftharpoons Ca^{2+} + C_2O_4^{2-}; K_{sp} = S^2$

$\therefore S = \sqrt{K_{sp}} = \sqrt{2.5 \times 10^{-9}} = 5 \times 10^{-5}$ mol/litre

solubility $= 5 \times 10^{-5} \times 128 = 0.0064$ g

Q.12 Answer:

$S = \{K_{sp}/6912\}^{1/7}$

Solution:

$[Zr_3(PO_4)_4] \rightleftharpoons 3Zr^{4+} + 4PO_4^{3-}$

$$K_{sp} = (3S)^3(4S)^4$$
$$= 27S^3 \times 256S^4$$
$$= 6912S^7$$
$$\therefore \quad S = \left(\frac{K_{sp}}{6912}\right)^{1/7}$$

Q.13 Answer:

5.1×10^{-5}M

Solution:

Given $Na_2CO_3 = 1.0 \times 10^{-4}$M

$\therefore \left[CO_3^{2-}\right] = 1.0 \times 10^{-4}$M

i.e. $s = 1.0 \times 10^{-4}$M

$$\left[Ba^{2+}\right]\left[CO_3^{2-}\right] = K_{sp} \text{ of } BaCO_3$$

At equilibrium $\left[Ba^{2+}\right] = \dfrac{K_{sp}}{\left[CO_3^{2-}\right]} = \dfrac{5.1 \times 10^{-9}}{1.0 \times 10^{-4}}$

$$= 5.1 \times 10^{-5}\text{M}$$

Q.14 Answer:

99.9

Solution:

To precipitate the $AgCl$

$\left[Ag^{+}\right]$ **required**

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$= \frac{K_{sp}(AgCl)}{[Cl^-]} = \frac{1.0\times10^{-10}}{0.1} = 1.0 \times 10^{-9}M$

$\left[Br^-\right]$ **left at this stage** $= \frac{K_{sp}(AgBr)}{[Ag^+]}$

$= \frac{1.0\times10^{-13}}{1.0\times10^{-9}} = 1.0 \times 10^{-4}M$

$\%$ **of remaining**

$\left[Br^-\right] = \frac{1.0\times10^{-4}}{0.1} \times 100 = 0.1$

$\%$ **of** Br^- **to be precipitated** $= 100 - 0.1 = 99.9$

Q.15 Answer:

$BaSO_4 > Hg_2Cl_2 > CrCl_3 > Cr_2(SO_4)_3$

Solution:

$Cr_2(SO_4)_3 \leftrightharpoons \underset{2s}{2Cr^{3+}} + \underset{3s}{3SO_4^{2-}}$

$K_{sp} = (2s)^2(3s)^3 = 4\,s^2 \times 27s^3 = 108s^5$

$s = \left(\frac{K_{sp}}{108}\right)^{1/5}$

$Hg_2Cl_2 \leftrightharpoons \underset{2s}{2Hg^{2+}} + \underset{2s}{2Cl^-}$

$K_{sp} = (2s)^2 \times (2s)^2 = 16s^4$

$s = \left(\frac{K_{sp}}{16}\right)^{1/4}$

$BaSO_4 \leftrightharpoons Ba^{2+} + SO_4^{2-}$

$K_{sp} = s^2$

$s = \sqrt{K_{sp}}$

$CrCl_3 \leftrightharpoons \underset{s}{Cr^{3+}} + 3Cl^-$

$K_{sp} = s \times (3s)^3 = 27s^4$

$s = \left(\frac{K_{sp}}{27}\right)^{1/4}$

Hence the correct order of solubilities of salts is

$\sqrt{K_{sp}} > \left(\frac{K_{sp}}{16}\right)^{1/4} > \left(\frac{K_{sp}}{27}\right)^{1/4} > \left(\frac{K_{sp}}{108}\right)^{1/5}$

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